# M.Tech Program

## Advanced Industry Integrated Programs

Jointly offered by University and LTI MindTree

# Graph RAG based Financial Insights

## Review 1

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# Introduction

## Problem

- Annual reports = **Tables + Text + Strategy + Risk**
- Financial information is **distributed across pages**
- Traditional search finds **text**, not relationships
- LLMs may generate **unsupported answers**

## Proposed Idea

- Financial Fact Extraction
- Entity & Relationship Extraction
- **Neo4j Knowledge Graph**
- Multi-Hop Graph Retrieval
- **Ollama + Local Model**
- Source/Page Grounding

# Objective

- Extract **financial facts**
- Identify **business entities**
- Discover **semantic relationships**
- Build a **financial knowledge graph**
- Perform **multi-hop reasoning**
- Retrieve **source/page evidence**
- Generate answers using **local model**
- Minimize **hallucination**

# Scope

## Input

- LTM Annual Reports
- Financial Statements
- Management Discussion
- Strategy & Risk
- Business Information

## Processing

- PDF Parsing
- Normalization
- Fact, Entity Extraction
- Relationship Extraction

## Knowledge

- Financial Metrics
- Financial Values
- Reporting Periods
- Source Pages

## Output

- Financial Answers
- Multi-Hop Reasoning
- Graph Evidence
- Page-Level Provenance

# Literature Review

| Paper | Key Contribution |
|---|---|
| Lewis et al. (2020) — RAG | External knowledge retrieval + LLM generation |
| Reimers & Gurevych (2019) — SBERT | Semantic embeddings for similarity search |
| Chen et al. (2021) — FinQA | Numerical reasoning over financial reports |
| Zhu et al. (2021) — TAT-QA | Question answering over financial text + tables |
| Edge et al. (2024) — GraphRAG | Knowledge graphs + multi-hop/global reasoning |

# Summary of Literature

## Key Findings

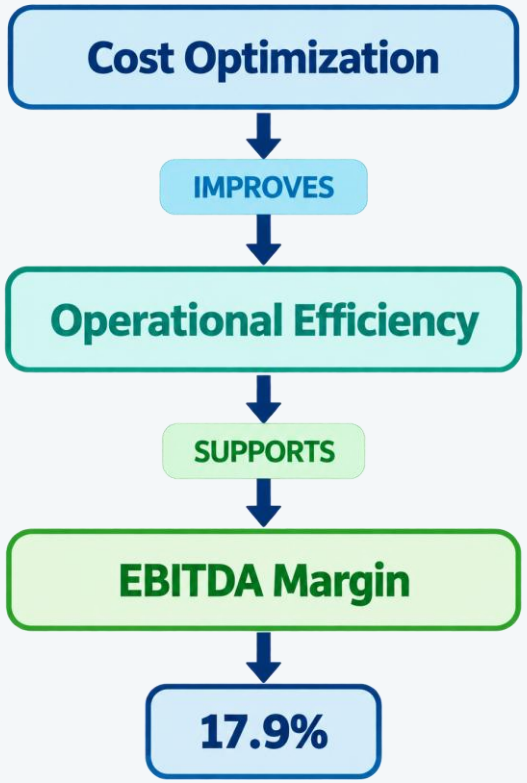
- **RAG** → External knowledge + LLM generation
- **SBERT** → Semantic similarity-based retrieval
- **FinQA** → Financial numerical reasoning
- **TAT-QA** → Text + table reasoning
- **GraphRAG** → Multi-hop relationship reasoning

## Research Gap

Limited integration of financial facts, graph relationships, document context, and source provenance.

# Additional Information

## How does cost optimization relate to EBITDA margin?




### Key Advantages

- **Relationship-Aware** Retrieval
- **Multi-Hop** Reasoning
- **Financial Fact** Retrieval
- **Source/Page** Traceability
- **Local LLM** Inference
- **Reduced Hallucination**
- **Explainable** Answers

# Modules to cover…. Calibri body -36

- **Data Ingestion** – PDF data extraction
- **Knowledge Graph** – Entity relationship mapping
- **Vector Database** – Semantic search storage
- **Hybrid Retrieval** – Graph + vector retrieval
- **Answer Generation** – LLM-powered insights
- **Multi-Year Analysis** – Fiscal trend tracking
- **Evaluation Module** – Accuracy assessment
- **User Interface** – Interactive dashboard
- **Infrastructure** – Deployment & orchestration

# Financial Intelligence GraphRAG Engine

- **Core Logic:** Decoupled LLM reasoning from financial math to eliminate hallucination.
- **Dual-DB Architecture:** Integrated Neo4j (structured facts) + Postgres/pgvector (narrative context).
- **Structured Parsing:** Deployed IBM Docling to preserve table integrity from complex PDF filings.
- **Intelligent Routing:** Developed hybrid_router.py to map user intent to optimal retrieval strategy.
- **LangGraph Orchestration:** Implemented stateful workflow to fuse graph nodes + vector chunks into verifiable answers.